

5주차 과제(A)

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C: → 소정님이 알려준 방향으로 코드 수정 및 연동 재실험

1. 메시지 파트의 품질 개선: `base64` 이벤트는 `plaintext_candidates` 에서 제외하고, 별도의 `base64_candidates` (또는 `encoded_candidates`) 리스트에 저장하는 방식으로 수정. (base64를 완전 제외하는 게 아님)
2. Python API에서 Java Hook이 안 걸리는 문제:

▼ import 파일 위치 수정(src. 추가)

```
python -m src.dynamic_analyzer.run_test
[1] frida-server 연결 시도...
    연결된 디바이스: Android Emulator 5554
[2] com.android.chrome spawn + attach...
[재시도 1/3] spawn 실패 (com.android.chrome): need Gadget to attach on jailed Android; its default location is: C:\Users\User\AppData\Local\Microsoft\Windows\INetCache\frida\gadget-android-arm64.so
[재시도 2/3] spawn 실패 (com.android.chrome): need Gadget to attach on jailed Android; its default location is: C:\Users\User\AppData\Local\Microsoft\Windows\INetCache\frida\gadget-android-arm64.so
[재시도 3/3] spawn 실패 (com.android.chrome): need Gadget to attach on jailed Android; its default location is: C:\Users\User\AppData\Local\Microsoft\Windows\INetCache\frida\gadget-android-arm64.so
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Users\User\OneDrive - 이화여자대학교\문서\GitHub\Android-X-Ray\src\dynamic_analyzer\run_test.py", line 49, in <module>
    main()
  File "C:\Users\User\OneDrive - 이화여자대학교\문서\GitHub\Android-X-Ray\src\dynamic_analyzer\run_test.py", line 23, in main
    pid = controller.spawn_and_attach(PACKAGE_NAME)
          ~~~~~
  File "C:\Users\User\OneDrive - 이화여자대학교\문서\GitHub\Android-X-Ray\src\dynamic_analyzer\frida_controller.py", line 45, in spawn_and_attach
    raise RuntimeError(f"{package_name} spawn {self.max_retries}회 실패: {last_error}")
RuntimeError: com.android.chrome spawn 3회 실패: need Gadget to attach on jailed Android; its default location is: C:\Users\User\AppData\Local\Microsoft\Windows\INetCache\frida\gadget-android-arm64.so
```

▼ 문제 해결을 위한 탐색

1. 에뮬레이터 이미지가 구글 play 버전인지
2. frida-server를 root로 잘 띄웠는지
3. frida-server 아키텍처 버전이 불일치인지

```
PS C:\Users\User> adb root
adb is already running as root
PS C:\Users\User> adb shell
generic_x86_64_arm64:/ # su
generic_x86_64_arm64:/ # cd /data/local/tmp
generic_x86_64_arm64:/data/local/tmp # ./frida-server &
[1] 12096
```

```
PS C:\Users\User> adb shell getprop ro.product.cpu.abi
x86_64
PS C:\Users\User> frida --version
17.9.1
PS C:\Users\User> adb shell /data/local/tmp/frida-server --version
17.9.1
```

→ 전부 아님. 그럼 남은 건? **SELinux가 Enforcing 모드**인 경우. (**jailed** 판정은 root 여부와 별개로, frida-server가 프로세스 주입에 필요한 헬퍼를 SELinux 정책 때문에 못 띄울 때도 뜸)

4. 'Enforcing'이 나오면 이게 문제다!
- adb shell 안에서 su 진입 후 확인:

```
adb shell
su
getenforce
```

```
C:\Users\User>adb shell
generic_x86_64_arm64:/ # su
generic_x86_64_arm64:/ # getenforce
Enforcing
```

[해결]

4-1. setenforce 0 해두고 frida-server 재시작

```
generic_x86_64_arm64:/ # setenforce 0
generic_x86_64_arm64:/ # getenforce
Permissive
```

```
generic_x86_64_arm64:/ # ps -A | grep frida
root      12096      1 1132528 133208 do_sys_poll
S frida-server
generic_x86_64_arm64:/ # kill 12096
generic_x86_64_arm64:/ # cd /data/local/tmp
generic_x86_64_arm64:/data/local/tmp # ./frida-server &
[1] 12223
```

[또다른 문제]

5. 이번엔 호환성 이슈. (그 이전 이슈가 맞음...)

```
[1] frida-server 연결 시도...
    연결된 디바이스: Android Emulator 5554
[2] com.android.chrome spawn + attach...
[재시도 1/3] spawn 실패 (com.android.chrome): unable to locate android.os.Process.set
ArgV0() slot; please file a bug
[재시도 2/3] spawn 실패 (com.android.chrome): unable to locate android.os.Process.set
ArgV0() slot; please file a bug
[재시도 3/3] spawn 실패 (com.android.chrome): unable to locate android.os.Process.set
ArgV0() slot; please file a bug
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
```

```
File "C:\Users\User\OneDrive - 이화여자대학교\문서\GitHub\Android-X-Ray\src\dynamic_analyzer\run_test.py", line 49, in <module>
    main()
File "C:\Users\User\OneDrive - 이화여자대학교\문서\GitHub\Android-X-Ray\src\dynamic_analyzer\run_test.py", line 23, in main
    pid = controller.spawn_and_attach(PACKAGE_NAME)
           ~~~~~^~~~~~
File "C:\Users\User\OneDrive - 이화여자대학교\문서\GitHub\Android-X-Ray\src\dynamic_analyzer\frida_controller.py", line 45, in spawn_and_attach
    raise RuntimeError(f"{package_name} spawn {self.max_retries}회 실패: {last_error}")
RuntimeError: com.android.chrome spawn 3회 실패: unable to locate android.os.Process.setArgV0() slot; please file a bug
```

- 상황 설명: ART가 이 메서드를 컴파일하는 방식이 기기/빌드마다 달라서, frida-gum의 패턴 매칭이 깨지는 케이스. API 30~35에 걸쳐서 폭넓게 보고되어 있고, 특정 최신 버전만의 문제가 아니라 여러 Android 버전에서 산발적으로 나타나는 고질적인 버그임. (πππ)
- 임시 해결: spawn 시 프로세스 이름을 앱 패키지명으로 리네이밍하는 단계 (setArgV0 호출)에서만 발생하기 때문에 앱을 미리 실행해두고 attach만 하면 이 코드 경로 자체를 거치지 않으니 괜찮을 것.

[추측 테스트]

6.

▼ 자바 브릿지 주입 실패 원인: 버전 안 맞음 (아직 못고침 - 가능한 버전이 애초에 달라서 찾는 중; 일까지)

frida(파이썬), frida-server, frida-tools 세 가지의 버전이 같지 않은 경우에 발생하는 문제

```
PS C:\Users\User> pip show Frida
Name: frida
Version: 17.16.4
Summary: Dynamic instrumentation toolkit for developers, reverse-engineers,
and security researchers
Home-page: https://frida.re
Author: Frida Developers
Author-email: olearv@frida.re
License: wxWindows Library Licence, Version 3.1
Location: C:\Users\User\AppData\Local\Programs\Python\Python312\Lib\site-pac
kages
Requires:
Required-by: Frida-tools

PS C:\Users\User> pip show frida-tools
Name: frida-tools
Version: 14.18.4
Summary: Frida CLI tools
Home-page: https://frida.re
Author: Frida Developers
Author-email: olearv@frida.re
License: wxWindows Library Licence, Version 3.1
Location: C:\Users\User\AppData\Local\Programs\Python\Python312\Lib\site-pac
kages
Requires: colorama, frida, prompt-toolkit, pygments, websockets
Required-by:
```

✓ A: adb shell tcpdump 실행/종료 subprocess 제어 초안

- [tcpdump-android-builds/build_x86_64.sh at master · extremecoders-re/tcpdump-android-builds · GitHub](#) (이미 컴파일된 바이너리를 제공)
- tcpdump 다운로드해야 함!!

```
# 1. 다운로드
curl -L -o tcpdump-x86_64.zip https://github.com/extremecoders-re/tcpdump-android-builds/files/2734355/tcpdump-x86_64.zip
Expand-Archive -Path tcpdump-x86_64.zip -DestinationPath .
# -> tcpdump-x86_64 바이너리 나눔

# 1-1. 확인하고 싶으면
dir tcpdump-x86 64
```

```
# 2. 기기로 push
adb push .\tcpdump-x86_64\tcpdump-x86_64 /data/local/tmp/tcpdump
adb shell chmod 755 /data/local/tmp/tcpdump

# 3. 확인
adb shell /data/local/tmp/tcpdump --version #4.9.2 나
```

- ☑ A: scenario_runner.py(D가 4주차에 만든 것)와 캡처 시작/종료 동기화
- ☑ A: 실제 capture.pcap 생성 확인 → B/C에게 전달

▼ 결과 (4주차와 똑같은 frida 호환성 문제, A 자체는 성공)

A가 만든 캡처 파이프라인은 완벽하게 동작했습니다.

증거:

- `pcap pull` 완료: `output\capture.pcap (5093 bytes)` → 실제 트래픽 캡처되고 pull까지 성공
- `CaptureMeta` 에 `package_name: com.google.android.calendar`, `capture_duration_sec: 8` 등 정확히 채워짐
- 크로스 패키지 연결(`network_analyzer` ↔ `dynamic_analyzer`)도 에러 없이 끝까지 실행됨

`Scenario success: False` 는 A 코드 문제가 아니라 D/B 쪽 `dynamic_analyzer` 시나리오 실행 자체가 실패한 것. `crashed: False` 인데 `success: False` 라는 건 `error` 필드에 뭔가 메시지가 들어있다는 뜻임

```
python -m src.network_analyzer.scenario_capture
<frozen runpy>:128: RuntimeWarning: 'src.network_analyzer.scenario_capture' found in
sys.modules after import of package 'src.network_analyzer', but prior to execution
of 'src.network_analyzer.scenario_capture'; this may result in unpredictable behaviour
[tcpdump_capture] $ adb get-state
[tcpdump_capture] $ adb shell getprop sys.boot_completed
[tcpdump_capture] 부팅 완료 확인 (0s 경과)
[tcpdump_capture] $ adb shell [ -f /data/local/tmp/tcpdump ] && echo EXISTS || echo
MISSING
[tcpdump_capture] 캡처 시작: adb shell /data/local/tmp/tcpdump -i any -w /data/local/
tmp/capture.pcap
[tcpdump_capture] $ adb shell pidof tcpdump
[tcpdump_capture] 캡처 시작 확인됨 (remote pid=14582)
[scenario_capture] flush 대기 3.0s...
[tcpdump_capture] $ adb shell kill -TERM 14582
[tcpdump_capture] 원격 tcpdump(pid=14582)에 SIGTERM 전송
[tcpdump_capture] 캡처 종료 완료
[tcpdump_capture] $ adb pull /data/local/tmp/capture.pcap output\capture.pcap
[tcpdump_capture] pcap pull 완료: output\capture.pcap (5093 bytes)
CaptureMeta: {'package_name': 'com.google.android.calendar', 'capture_started_at':
'2026-07-25T05:47:19+00:00', 'capture_duration_sec': 6, 'pcap_file': 'output\capture.pcap'}
Scenario success: False
Scenario crashed: False
Scenario error: [Errno 2] No such file or directory: 'dynamic_analyzer/hooks.bundle.js'
B/C에게 전달 준비 완료 -> dns_parser.py / sni_parser.py 입력으로 사용 가능
```

- 수정 후 재시도 결과:

```
Scenario success: True
Scenario crashed: False
Scenario error: None
```

- 재수정 후 결과: Frida 후킹이 실제 작동함
 - base64 혹은 encodeToString/decode를 여러 번 잡아냄. (MIE...로 시작하는 게 Base64 문자열들. = X.509인증서)
 - hookStringBuilder, hookCipher도 등록 됐는데 조건에 안 걸려서 지금은 안 잡힘.
 - 혹 자체는 다 살아있음.

▼ 코드

```

MKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNTW91bnRh\NaW4gVmllZEqMA4GA1UEChMHQW5kcm9pZDEQMA4GA1UEC
xMHQW5kcm9pZDEQMA4GA1UEAxMHQW5k\ncm9pZDEiMCAGCSqGSIB3DQEJARYTYW5kcm9pZEBhbmRyb2
lkLmNvbYIJanWFuGx90071MAwGA1Ud\NEwQFMAMBAf8wDQYJKoZIhvcNAQEEBQADggEBABnTDEPF+3i
SP0wNfdIjIz1AlnrPzgAIHVvXxunW\N7SBrDhEglQZBbKJEk5kT0mtKo0D1JMrSu1xuTKEBahWRbqHsXc
laXjoBADb0kkjVEJu/Lh5hgYZn\N0jvLba8Ld7HCKePCVePoTJBdI4fvugnL8TsgK05aIskyY0hKI9L8K
fqfGTL1lZ0v2KoWD0KWwtAW\NpOGChZxmQ+nBli+gWYMz1vAkP+aayLe0a1EQiml0a10762r0GX00ks+
UeXde2Z4e+8S/pf7pITE\NI/tP+MxJtALw9QUWEv9lKtk+jkbqxbsh8nfBUapfKqYn0eidpwq2AzVp3ju
Yl7//fKnaPhJD9gs=\n', 'extra': {'direction': 'encode'}, 'thread_id': 14841}}
[디버그] 전체 메시지: {'type': 'send', 'payload': {'hook_type': 'base64', 'timestam
p': '2026-07-25T05:55:58.784Z', 'class_name': 'android.util.Base64', 'method_nam
e': 'encodeToString', 'raw_value': 'MIIeqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSIB3D
QEBBAUAMIGUMQswCQYDVQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNTW91bnRh
aW4gVmllZEqMA4GA1UEChMHQW5k\ncm9pZDEQMA4GA1UECxMHQW5kcm9pZDEQMA4GA1UEAxMHQW5kcm9
pZDEiMCAGCSqGSIB3DQEJARYT\NYW5kcm9pZEBhbmRyb2lkLmNvbTAeFw0wODA0MTUyMzY2NTZaFw0zNT
A5MDEyMzY2NTZaMIGUMQsw\NCQYDVQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNT
W91bnRhaw4gVmllZEq\NMA4GA1UEChMHQW5kcm9pZDEQMA4GA1UECxMHQW5kcm9pZDEQMA4GA1UEAxMH
QW5kcm9pZDEiMCAG\NCSqGSIB3DQEJARYTYW5kcm9pZEBhbmRyb2lkLmNvbTCCASAwDQYJKoZIhvcNAQE
BBQADggENADCC\NAQgCggEBANb0LggKv+IxTdGns8/TGFy0PTP6DHTHvbbR24kT9ixc0d9W+EaBPWW+wP
PKQmsHxajt\NwJmQwWfna8mZuSeJS48LIgAZlKkpFeVyxW0qMBujb8X8ETrWy550NaftI6t9+u7hZeTfH
wqNvacK\Nhp1RbE6dBRGWynwMVX8XW8N1+UjFaq6GCJukT4qmpN2afb8sCjUigq0GuMwYXrFVee74bQg
LHWGJ\NwPvmLHC69EH6kWr22ijx40KXlSIx2xT1AsSHee70w5iDBiK4aph27yH3TxkXy9V89TDdexAcK
k/\NcVHYNnDBapcavl7y0RiQ4biu8ymM8Ga/nmzhRKya6G0cGw8CAQ0jgfwgfkWQYDVR00BBYEFi0c
\Nxb6VTEM8YYY6FbBMvAPyT+CyMIHJBgNVHSMGcgEwgb6AFi0cxb6VTEM8YYY6FbBMvAPyT+CyoYGa\Np
IGXMIGUMQswCQYDVQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNTW91bnRh\NaW4g
VmllZEqMA4GA1UEChMHQW5kcm9pZDEQMA4GA1UECxMHQW5kcm9pZDEQMA4GA1UEAxMHQW5k\ncm9pZDE
iMCAGCSqGSIB3DQEJARYTYW5kcm9pZEBhbmRyb2lkLmNvbYIJanWFuGx90071MAwGA1Ud\NEwQFMAMBAf
8wDQYJKoZIhvcNAQEEBQADggEBABnTDEPF+3iSP0wNfdIjIz1AlnrPzgAIHVvXxunW\N7SBrDhEglQZBb
KJEk5kT0mtKo0D1JMrSu1xuTKEBahWRbqHsXclaXjoBADb0kkjVEJu/Lh5hgYZn\N0jvLba8Ld7HCKePC
VePoTJBdI4fvugnL8TsgK05aIskyY0hKI9L8KfqfGTL1lZ0v2KoWD0KWwtAW\NpOGChZxmQ+nBli+gWYM
z1vAkP+aayLe0a1EQiml0a10762r0GX00ks+UeXde2Z4e+8S/pf7pITE\NI/tP+MxJtALw9QUWEv9lKt
k+jkbqxbsh8nfBUapfKqYn0eidpwq2AzVp3juYl7//fKnaPhJD9gs=\n', 'extra': {'direction':
'encode'}, 'thread_id': 14841}}
[base64] encodeToString → MIIeqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSIB3DQEBBAUAMIG
UMQsw
[스크립트 메시지] {'type': 'send', 'payload': {'hook_type': 'base64', 'timestamp':
'2026-07-25T05:55:58.784Z', 'class_name': 'android.util.Base64', 'method_name':
'decode', 'raw_value': 'MIIeqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSIB3DQEBBAUAMIGUM
QswCQYDVQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNTW91bnRhaw4gVmllZEq
MA4GA1UEChMHQW5k\ncm9pZDEQMA4GA1UECxMHQW5kcm9pZDEQMA4GA1UEAxMHQW5kcm9pZDEiMCAGCSq
GSIB3DQEJARYT\NYW5kcm9pZEBhbmRyb2lkLmNvbTAeFw0wODA0MTUyMzY2NTZaFw0zNTA5MDEyMzY2NT
ZaMIGUMQsw\NCQYDVQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNTW91bnRhaw4gV
mllZEq\NMA4GA1UEChMHQW5kcm9pZDEQMA4GA1UECxMHQW5kcm9pZDEQMA4GA1UEAxMHQW5kcm9pZDEi
MCAG\NCSqGSIB3DQEJARYTYW5kcm9pZEBhbmRyb2lkLmNvbTCCASAwDQYJKoZIhvcNAQEEBQADggENADC
\NAQgCggEBANb0LggKv+IxTdGns8/TGFy0PTP6DHTHvbbR24kT9ixc0d9W+EaBPWW+wPPKQmsHxajt\N
wJmQwWfna8mZuSeJS48LIgAZlKkpFeVyxW0qMBujb8X8ETrWy550NaftI6t9+u7hZeTfHwqNvacK\Nhp1
RbE6dBRGWynwMVX8XW8N1+UjFaq6GCJukT4qmpN2afb8sCjUigq0GuMwYXrFVee74bQgLHWGJ\NwPvmL
HC69EH6kWr22ijx40KXlSIx2xT1AsSHee70w5iDBiK4aph27yH3TxkXy9V89TDdexAcKk/\NcVHYNnDBa
pcavl7y0RiQ4biu8ymM8Ga/nmzhRKya6G0cGw8CAQ0jgfwgfkWQYDVR00BBYEFi0c\Nxb6VTEM8YYY6
FbBMvAPyT+CyMIHJBgNVHSMGcgEwgb6AFi0cxb6VTEM8YYY6FbBMvAPyT+CyoYGa\NpIGXMIGUMQswCQY
DVQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcM5pYTEWMBQGA1UEBxMNTW91bnRh\NaW4gVmllZEqMA4GA1
UEChMHQW5kcm9pZDEQMA4GA1UECxMHQW5kcm9pZDEQMA4GA1UEAxMHQW5k\ncm9pZDEiMCAGCSqGSIB3D
QEJARYTYW5kcm9pZEBhbmRyb2lkLmNvbYIJanWFuGx90071MAwGA1Ud\NEwQFMAMBAf8wDQYJKoZIhvcN
AQEEBQADggEBABnTDEPF+3iSP0wNfdIjIz1AlnrPzgAIHVvXxunW\N7SBrDhEglQZBbKJEk5kT0mtKo0D

```

```
1JMrSu1xuTKEBahWRbqHsXclaxjoBADb0kkjVEJu/Lh5hgYZn\n0jvLba8Ld7HCKePCVePoTJBdI4fvug  
nL8TsgK05aIskyY0hKI9L8KfqfGTL1lZ0v2KoWd0KwWtAW\nPoGChZxmQ+nBli+gWYMzM1vAkP+aayLe0  
a1EQiml0a10762r0GX00ks+UeXde2Z4e+8S/pf7pITE\nI/tP+MxJtALw9QUUEv9lKtk+jkbqxbsh8nfB  
UapfKqYn0eidpwq2AzVp3juYl7//fKnaPhJD9gs=\n', 'extra': {'direction': 'decode'}, 't  
hread_id': 14841}}
```

```
[디버그] 전체 메시지: {'type': 'send', 'payload': {'hook_type': 'base64', 'timestam  
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IhvcNAQEEBQADggEBABnTDPEF+3iSP0wNfdIjIz1A\nlrPzgAIHVVxXunW\n7SBRdHEglQZBbKJEk5kT0m
tKoOD1JMrSu1xuTKEBahWRbqHsXclaxjoBADb0kkjVEJu/Lh5hgYZn\n0jvLba8Ld7HCKePCVePoTJBdI
4fvugnL8TsgK05aIskyY0hKI9L8KfqfGTl1lZ0v2KoWD0KwWtAW\nPoGChZxmQ+nBli+gWYMzM1vAkP+a
ayLe0a1EQiml0a10762r0GX00ks+UeXde2Z4e+8S/pf7pITE\nI/tP+MxJtALw9QUUEv9lKtk+jkbqxb
sh8nfBUapfKqYn0eidpwq2AzVp3juYl7//fKnaPhJD9gs=\n', 'extra': {'direction': 'decod
e'}, 'thread_id': 14841}}
[base64] decode → MIIeqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSib3DQEBBAUAMIGUMQsw

```

```
[스크립트 메시지] {'type': 'send', 'payload': {'hook_type': 'base64', 'timestamp': '2026-07-25T05:55:58.806Z', 'class_name': 'android.util.Base64', 'method_name': 'encodeToString', 'raw_value': 'MIIEQzCCAyugAwIBAgIJAMLgh0ZkSjCNMA0GCSqGSIb3DQEBB
AUAMHqxCzAJBgNVBAYTALVTMRMw\neQYDVQIQIEwpDYWxpZm9ybmlhMRYwFAYDVQQHEw1Nb3VudGFpbW
aWV3MRQwEgYDVQKKEwtHb29n\ncGUgSW5jLjEQMA4GA1UECXMHQW5kcm9pZDEQMA4GA1UEAxMHQW5kcm9
pZDAeFw0wODA4MjEyMzEz\nMzRaFw0zNjAxMDcyMzEzMzRaMHQxCzAJBgNVBAYTALVTMRMwEQYDVQIQIEw
pDYWxpZm9ybmlhMRYw\neFAYDVQQHEw1Nb3VudGFpbWwV3MRQwEgYDVQKKEwtHb29nbnGUgSW5jLjEQM
A4GA1UECXMHQW5k\ncm9pZDEQMA4GA1UEAxMHQW5kcm9pZDCCASAwDQYJKoZIhvcNAQEBBQADggENADCC
AQgCggEBAKtW\nLgDY06IIrgqWbxJ0KdoR8qtW0I9Y4sypEwPpt1TTcvZApxsdyxMJZ2J0Rland2qSGT2
y5b+3JKKe\ndxILdmpHpDsz2WCbdxgxRczfey5YznTJ4VZbH0xqWVW/8lGmPav5xVwnIiJS6HXK+BVKZF
+JcwJA\nsb/GEuq/eFdpuzSqeYTcfi6idkyugwfYwXFU1+5fZKUaRKYCwkkFQVfcAs1fXA5V+++FGfvj
J/Cx\nuRaSxaBvGdGDhfXE28LWuT9ozCl5xw4Yq50GazvV24mZVSo000yZ31j7kYvtwYK6NeAdwbSxDdJ
E\nq04k//0z0HKRUiGYxtqw/A0LFFtqoZKFjnkCAQ0jgdkwgdYwHQYDVR00BBYEFMd9jMIhF1Ylmn/T\n
gt9r45jk14aLMIGmBgNVHSMegZ4wgZuAFMd9jMIhF1Ylmn/Tgt9r45jk14aLoXikdjB0MQswCQYD\n
nVQQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcn5pYTEwMBQGA1UEBxMNTW91bnRhaW4gVmlldzEUMBIG\n
nA1UEChMLR29vZ2x1IEluYy4xEDA0BgNVBAsTB0FuZlJvaWwQxEDA0BgNVBAMTB0FuZlJvaWwSCCQDC\n
n4IdGZEowjTAMBgNVHRMEBTADAQH/MA0GCSqGSIb3DQEBBAAUAA4IBAQBt0LL074UwLDYKqs6Tm8/y\n
nzKkEu116FmH4rkaymUIE0P9KaMftGLMexFlaYjzmB20xZyl6euNXEsQH8gjwyxCUKRJNexBiGcCE\n
nyj6z+a1fuHHvkiaai+KL8W1EyNmgjmyy8AW7P+LLlkr+ho5zEHatRbM/YAnqGcFh5iZBqpkHf1S\n
\nkMXFh4dd239FJ1jWYfbMDMy3NS5CTMQ2XFI1MvcyUTdZPERjQfTbQe3aDQsQcafEQPD+nqActifK\n
\nZ0Np0IS9L9kR/wbNvyz6ENwPiTrjV2KRKEjH78ZMcUQXg0L3BYHJ3lc69Vs5Ddf9uUGGMYldX3Wf\n
\nMBEmh/9iFBDAaTCK\n', 'extra': {'direction': 'encode'}, 'thread_id': 14841}}
```

```
[디버그] 전체 메시지: {'type': 'send', 'payload': {'hook_type': 'base64', 'timestamp': '2026-07-25T05:55:58.806Z', 'class_name': 'android.util.Base64', 'method_name': 'encodeToString', 'raw_value': 'MIIEQzCCAyugAwIBAgIJAMLgh0ZkSjCNMA0GCSqGSIb3DQEBB
AUAMHqxCzAJBgNVBAYTALVTMRMw\neQYDVQIQIEwpDYWxpZm9ybmlhMRYwFAYDVQQHEw1Nb3VudGFp
biBWaWV3MRQwEgYDVQKKEwtHb29n\ncGUgSW5jLjEQMA4GA1UECXMHQW5kcm9pZDEQMA4GA1UEAxMHQW5
kcm9pZDAeFw0wODA4MjEyMzEz\nMzRaFw0zNjAxMDcyMzEzMzRaMHQxCzAJBgNVBAYTALVTMRMwEQYDVQ
QIEwpDYWxpZm9ybmlhMRYw\neFAYDVQQHEw1Nb3VudGFpbWwV3MRQwEgYDVQKKEwtHb29nbnGUgSW5jL
jEQMA4GA1UECXMHQW5k\ncm9pZDEQMA4GA1UEAxMHQW5kcm9pZDCCASAwDQYJKoZIhvcNAQEBBQADggEN
ADCCAQgCggEBAKtW\nLgDY06IIrgqWbxJ0KdoR8qtW0I9Y4sypEwPpt1TTcvZApxsdyxMJZ2J0Rland2q
SGT2y5b+3JKKe\ndxILdmpHpDsz2WCbdxgxRczfey5YznTJ4VZbH0xqWVW/8lGmPav5xVwnIiJS6HXK+B
VKZF+JcwJA\nsb/GEuq/eFdpuzSqeYTcfi6idkyugwfYwXFU1+5fZKUaRKYCwkkFQVfcAs1fXA5V+++FG
fvjJ/Cx\nuRaSxaBvGdGDhfXE28LWuT9ozCl5xw4Yq50GazvV24mZVSo000yZ31j7kYvtwYK6NeAdwbSx
DdJE\nq04k//0z0HKRUiGYxtqw/A0LFFtqoZKFjnkCAQ0jgdkwgdYwHQYDVR00BBYEFMd9jMIhF1Ylmn/
T\ngt9r45jk14aLMIGmBgNVHSMegZ4wgZuAFMd9jMIhF1Ylmn/Tgt9r45jk14aLoXikdjB0MQswCQYD\n
nVQQGEwJVUzETMBEGA1UECBMKQ2FsaWZvcn5pYTEwMBQGA1UEBxMNTW91bnRhaW4gVmlldzEUMBIG\n
nA1UEChMLR29vZ2x1IEluYy4xEDA0BgNVBAsTB0FuZlJvaWwQxEDA0BgNVBAMTB0FuZlJvaWwSCCQDC\n
n4IdGZEowjTAMBgNVHRMEBTADAQH/MA0GCSqGSIb3DQEBBAAUAA4IBAQBt0LL074UwLDYKqs6Tm8/y\n
nzKkEu116FmH4rkaymUIE0P9KaMftGLMexFlaYjzmB20xZyl6euNXEsQH8gjwyxCUKRJNexBiGcCE\n
nyj6z+a1fuHHvkiaai+KL8W1EyNmgjmyy8AW7P+LLlkr+ho5zEHatRbM/YAnqGcFh5iZBqpkHf1S\n
\nkMXFh4dd239FJ1jWYfbMDMy3NS5CTMQ2XFI1MvcyUTdZPERjQfTbQe3aDQsQcafEQPD+nqActifK\n
\nZ0Np0IS9L9kR/wbNvyz6ENwPiTrjV2KRKEjH78ZMcUQXg0L3BYHJ3lc69Vs5Ddf9uUGGMYldX3Wf\n
\nMBEmh/9iFBDAaTCK\n', 'extra': {'direction': 'encode'}, 'thread_id': 14841}}
```

[base64] encodeToString → MIIEQzCCAyugAwIBAgIJAMLgh0ZkSjCNMA0GCSqGSIb3DQEBBAAUAMHqxCzAJ

```
[스크립트 메시지] {'type': 'send', 'payload': {'hook_type': 'base64', 'timestamp': '2026-07-25T05:55:58.807Z', 'class_name': 'android.util.Base64', 'method_name': 'decode', 'raw_value': 'MIIEQzCCAyugAwIBAgIJAMLgh0ZkSjCNMA0GCSqGSIb3DQEBB
AUAMHqxCzAJBgNVBAYTALVTMRMw\neQYDVQIQIEwpDYWxpZm9ybmlhMRYwFAYDVQQHEw1Nb3VudGFpbW
aWV3MRQwEgYDVQKKEwtHb29n\ncGUgSW5jLjEQMA4GA1UECXMHQW5kcm9pZDEQMA4GA1UEAxMHQW5kcm9pZDAeFw0
wODA4MjEyMzEz\nMzRaFw0zNjAxMDcyMzEzMzRaMHQxCzAJBgNVBAYTALVTMRMwEQYDVQIQIEwpDYWxpZm
9ybmlhMRYw\neFAYDVQQHEw1Nb3VudGFpbWwV3MRQwEgYDVQKKEwtHb29nbnGUgSW5jLjEQMA4GA1UEC
XMHQW5k\ncm9pZDEQMA4GA1UEAxMHQW5kcm9pZDCCASAwDQYJKoZIhvcNAQEBBQADggENADCCAQgCggE
BAKtW\nLgDY06IIrgqWbxJ0KdoR8qtW0I9Y4sypEwPpt1TTcvZApxsdyxMJZ2J0Rland2qSGT2y5b+3JKK
```

```
e\ndxiLDmpHpDsz2WCbdxgxRczfey5YZnTJ4VZbH0xqWVW/8lGmPav5xVwnIiJS6HXk+BVKZF+JcWjA\n
sb/GEuq/eFdpuzSqeYTcfi6idkyugwfYwXFU1+5fZKUaRKYCwkkFQVfcAs1fXA5V+++FGfvjJ/Cx\nURa
SxaBvGdGDhfXE28LWuT9ozCL5xw4Yq50GazvV24mZVSo000yZ31j7kYvtwYK6NeAdwbSxDdJE\nq04k//
0z0HKrUiGYXtqw/A0LFFtqoZKFjnkCAQ0jgdkwgdYwHQYDVR00BBYEFMD9jMIhF1Ylmn/T\ngt9r45jk1
4aIMIGmBgNVHSMegZ4wgZuAFMD9jMIhF1Ylmn/Tgt9r45jk14aloXikdjB0MQswCQYD\nnVQQGEwJVUzET
MBEGA1UECBMKQ2FsaWZvcn5pYTEwMBQGA1UEBxMNTW91bnRhaW4gVmllZDZUMBIG\nnA1UEChMLR29vZ2x
lIEluYy4xEDA0BgNVBAsTB0FuZHZJvaWQxEDA0BgNVBAMTB0FuZHZJvaWSSCCQDC\nn4IdGZEowjTAMBgNVHR
MEBTADAQH/MA0GCSqGSIb3DQEBAUAA4IBAQBt0LL074UwLDYKqs6Tm8/y\nnzKkEu116FmH4rkaymUIE0
P9KaMftGLMexFlaYjzmB20xZyl6euNXEsQH8gjwyxCUKRJNexBiGcCE\nnyj6z+a1fuHHvkiaai+KL8W1E
yNmgjmyy8AW7P+LLlkr+ho5zEHatRbM/YAnqGcFh5iZBqpkHf1S\nnKMXFh4dd239FJ1jWYfbMDMy3NS5
CTMQ2XFII1MvcyUTdZPERjQfTbQe3aDQsQcaFEQPD+nqActifK\nnZ0Np0IS9L9kR/wbNvyz6ENwPiTrjV2
KRKEjH78ZMcUQXg0L3BYHJ3lc69Vs5Ddf9uUGGMYldX3Wf\nnMBEmh/9iFBDAaTCK\n', 'extra': {'d
irection': 'decode'}, 'thread_id': 14841}}
[디버그] 전체 메시지: {'type': 'send', 'payload': {'hook_type': 'base64', 'timestam
p': '2026-07-25T05:55:58.807Z', 'class_name': 'android.util.Base64', 'method_nam
e': 'decode', 'raw_value': 'MIIEQzCCAyugAwIBAgIJAMLgh0ZkSjCNMA0GCSqGSIb3DQEBAUAM
HQxCzAJBgNVBAYTA1VTMRMw\nnEQYDVQIEwPDYXpZm9ybmlhMRYwFAYDVQQHEw1Nb3VudGFpbiBwaWV3
MRQwEgYDVQKKEwHb29n\nnbGUgSW5jLjEQA4GA1UECXMHQW5kcm9pZDEQMA4GA1UEAxMHQW5kcm9pZDA
eFw0w0DA4MjEyMzEz\nnMzRaFw0zNjAxMDcyMzEzRaMHQxCzAJBgNVBAYTA1VTMRMwEQYDVQKKEwPDYXpZm9ybmlhMRYw\nnFAYDVQQHEw1Nb3VudGFpbiBwaWV3MRQwEgYDVQKKEwHb29n\nnbGUgSW5jLjEQA4GA
1UECXMHQW5k\nncm9pZDEQMA4GA1UEAxMHQW5kcm9pZDCCASAwDQYJKoZIhvcNAQEBBQADggENADCCAQgC
ggEBAKtW\nnLgDY06I1rgqWbxJ0KdoR8qtW0I9Y4sypEwPpt1TTcvZApxsdyxMJ22J0Rland2qSGT2y5b+
3JKke\nndxiLDmpHpDsz2WCbdxgxRczfey5YZnTJ4VZbH0xqWVW/8lGmPav5xVwnIiJS6HXk+BVKZF+JcW
jA\nnsb/GEuq/eFdpuzSqeYTcfi6idkyugwfYwXFU1+5fZKUaRKYCwkkFQVfcAs1fXA5V+++FGfvjJ/Cx
\nURaSxaBvGdGDhfXE28LWuT9ozCL5xw4Yq50GazvV24mZVSo000yZ31j7kYvtwYK6NeAdwbSxDdJE\nnq
04k//0z0HKrUiGYXtqw/A0LFFtqoZKFjnkCAQ0jgdkwgdYwHQYDVR00BBYEFMD9jMIhF1Ylmn/T\ngt9r
45jk14aIMIGmBgNVHSMegZ4wgZuAFMD9jMIhF1Ylmn/Tgt9r45jk14aloXikdjB0MQswCQYD\nnVQQGEwJ
VUzETMBEGA1UECBMKQ2FsaWZvcn5pYTEwMBQGA1UEBxMNTW91bnRhaW4gVmllZDZUMBIG\nnA1UEChMLR2
9vZ2x1IEluYy4xEDA0BgNVBAsTB0FuZHZJvaWQxEDA0BgNVBAMTB0FuZHZJvaWSSCCQDC\nn4IdGZEowjTAM
BgNVHRMEBTADAQH/MA0GCSqGSIb3DQEBAUAA4IBAQBt0LL074UwLDYKqs6Tm8/y\nnzKkEu116FmH4rkay
mUIE0P9KaMftGLMexFlaYjzmB20xZyl6euNXEsQH8gjwyxCUKRJNexBiGcCE\nnyj6z+a1fuHHvkiaai+K
L8W1EyNmgjmyy8AW7P+LLlkr+ho5zEHatRbM/YAnqGcFh5iZBqpkHf1S\nnKMXFh4dd239FJ1jWYfbMDM
y3NS5CTMQ2XFII1MvcyUTdZPERjQfTbQe3aDQsQcaFEQPD+nqActifK\nnZ0Np0IS9L9kR/wbNvyz6ENwPi
TrjV2KRKEjH78ZMcUQXg0L3BYHJ3lc69Vs5Ddf9uUGGMYldX3Wf\nnMBEmh/9iFBDAaTCK\n', 'extr
a': {'direction': 'decode'}, 'thread_id': 14841}}
[base64] decode → MIIEQzCCAyugAwIBAgIJAMLgh0ZkSjCNMA0GCSqGSIb3DQEBAUAMHQxCzAJ
[scenario_capture] flush 대기 3.0s...
[tcpdump_capture] $ adb shell kill -TERM 14826
[tcpdump_capture] 원격 tcpdump(pid=14826)에 SIGTERM 전송
[tcpdump_capture] 캡처 종료 완료
[tcpdump_capture] $ adb pull /data/local/tmp/capture.pcap output\capture.pcap
[tcpdump_capture] pcap pull 완료: output\capture.pcap (613353 bytes)
CaptureMeta: {'package_name': 'com.google.android.calendar', 'capture_started_a
t': '2026-07-25T05:55:57+00:00', 'capture_duration_sec': 15, 'pcap_file': 'output
\capture.pcap'}
Scenario success: True
Scenario crashed: False
Scenario error: None
B/C에게 전달 준비 완료 -> dns_parser.py / sni_parser.py 입력으로 사용 가능
```

- 결과 해석:
 - 613KB짜리 실제 pcap 캡처됨
 - CaptureMeta도 schema 그대로 채워짐

- success: True (Frida 혹은 adb 시나리오 + 캡처 동기화, 전체 파이프라인이 예외 없이 끝까지 돌았음)

```
pcap pull 완료: output\capture.pcap (613353 bytes)  
CaptureMeta: {..., 'capture_duration_sec': 15, 'pcap_file': 'output\\capture.pcap'}
```


▼ 패킷은 있는데 DNS/TLS: 0 문제

```
python -m src.network_analyzer.verify_and_handoff output/capture.pcap
{
  "pcap_path": "C:\\Users\\User\\OneDrive - 이화여자대학교\\문서\\GitHub\\Android-X-Ray\\output\\capture.pcap",
  "pcap_size_bytes": 613353,
  "verified_at": "2026-07-25T16:04:06",
  "packet_count": 390,
  "dns_query_count": 0,
  "dns_query_sample": [],
  "tls_sni_count": 0,
  "tls_sni_sample": [],
  "capture_summary": "=====\n| IO Statistics
|\n|
|\n| Duration: 9.554 secs |\n| In
terval: 9.554 secs |\n| Col 1: F
rames and bytes |\n|-----|\n|
1 |\n| Interval | Frames | Bytes |\n|-----
-----|\n| 0.000 <> 9.554 | 390 | 607089 |\n=====
====",
  "protocol_hierarchy": "=====\nProtocol Hierarchy Statistics\nFilter: \n\nframe
frames:390 bytes:607089\n sll frames:390 bytes:
607089\n ip frames:389 bytes:606977\n tcp
frames:389 bytes:606977\n http frames:2 bytes:48
0\n websocket frames:13 bytes:2361\n data
frames:13 bytes:2361\n data frames:176 bytes:590
640\n ipv6 frames:1 bytes:112\n icmpv6
frames:1 bytes:112\n=====
====",
  "notes": [
    "[WARNING] DNS 쿼리가 없습니다. DoH/DoT 사용 또는 캡처 인터페이스 문제일 수 있습니다.",
    "[WARNING] TLS ClientHello가 없습니다. TLS 재사용(Session Resume), ECH 또는 평문 통
신 가능성을 확인하세요.",
    "[WARNING] 패킷은 존재하지만 DNS/TLS가 모두 없습니다. protocol_hierarchy를 확인하여 실
제 프로토콜을 분석하세요."
  ]
}
인수인계 문서 작성 완료: ./HANDOFF_to_B_C.md
```

▼ 결과 해석: tcpdump 정상 동작. (캡처 실패 아님)

```
패킷 수      : 390개
캡처 시간    : 9.5초
크기         : 607KB
```

*프로토콜 계층에 TLS가 없음

```
ip
├─ tcp
│   ├── http (2)
│   ├── websocket (13)
│   └─ data (176)
```


▼ 가능성 높은 3가지

1. tcpdump가 앱 트래픽이 아니라 다른 TCP만 잡고 있다.
 - a. ADB, WebSocket, Chrome DevTools, Frida
 2. 이미 TLS handshake가 끝난 뒤부터 캡처했다.
 - a. 앱 실행 → tls handshake → 캡처 시작 (이러면 ClientHello가 없음)
 3. TLS인데 Wireshark가 TLS로 인식하지 못한다.
 - a. non-standard port일 경우
- 데이터 패킷 분석
 - 모든 통신이 127.0.0.1(localhost)
 - 즉, 앱이 외부 서버와 통신한 것이 아니라 단말 내부(loopback)에서 이루어진 통신만 캡처되고 있

```
tshark -r output/capture.pcap -q -z conv,tcp
=====
TCP Conversations
Filter:<No Filter>


```

Start	End	Duration	Protocol	Source	Destination	Frames	Bytes	Total	Re
127.0.0.1:60541	127.0.0.1:40905	8.6247	TCP	127.0.0.1:60541	127.0.0.1:40905	179	91 kB	179 511 kB	358 603 kB
127.0.0.1:43845	127.0.0.1:27042	0.000000000	TCP	127.0.0.1:43845	127.0.0.1:27042	14	1708 bytes	17 2237 bytes	31 3945 bytes

```
=====
```

☐ A: 캡처 실패/권한 문제 예외처리, 시간 제한 옵션